

Construction of extremely low-symmetry double-stranded helicates based on terpyridine ligands with consecutive unsymmetrical modification

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The development of artificial supramolecular double-stranded helical structures has received widespread attention; however, the reports focusing on the construction and resolution of double-stranded helical assemblies based on terpyridine are relatively scarce. Herein, we report a series of extremely low-symmetry double-stranded helicates (S^3 , S^4 , and S^{RS}) based on the head-to-tail coordination mode of ladder-style ligands (L^3 , L^4 , and L^{RS}), which are apparently different from the conventional helicates with symmetrical axis. The ladder-style tridentate ligand L^3 was first designed and synthesized by characteristic consecutive unsymmetrical modification of terpyridine. The chiral group 2,6-bis(oxazolonyl)pyridine (PyBox) was designed at the tail of tridentate ligand L^3 and achieved the chiral resolution of the assembly. Moreover, the self-assembly of mixed three ladder-style ligands (didentate L^2 , tridentate L^3 , and tetradentate L^4) and single terpyridine ligand (L^1) with Zn(II) also exhibited excellent narcissistic self-sorting behavior, without any statistical mixture.

double-stranded helicate, terpyridine, unsymmetrical modification, self-sorting behavior

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1 Introduction

The helix is one of the most fundamental structures in biological macromolecules such as α -helical protein (single helix), DNA (double helix), and collagen (triple helix), which are spontaneously formed from self-assembly and are closely linked to their exquisite biological functions. Inspired by biologically occurring phenomena in nature, chemists constructed many artificial multiple-stranded helical structures assisted by non-covalent interactions and achieved functions based on simple building blocks [1,2]. Among

various non-covalent interactions, coordination-driven self-assembly has been proven to be one of the most powerful bottom-up approaches for constructing well-defined supramolecular architectures, due to its highly directional and predictable feature [3–6]. Since the term helicate was introduced by Lehn and co-workers [7] in 1987, describing of self-assemblies with two or more metal atoms linked by short or relatively rigid multidentate organic ligands, many excellent works regarding the coordination-driven self-assembly, structures and properties of helicates have been reported over the past decades [8–12]. In these successful examples, not only high complementarity and affinity for the ligands to bind metals but also appropriate linkers for the forming of a

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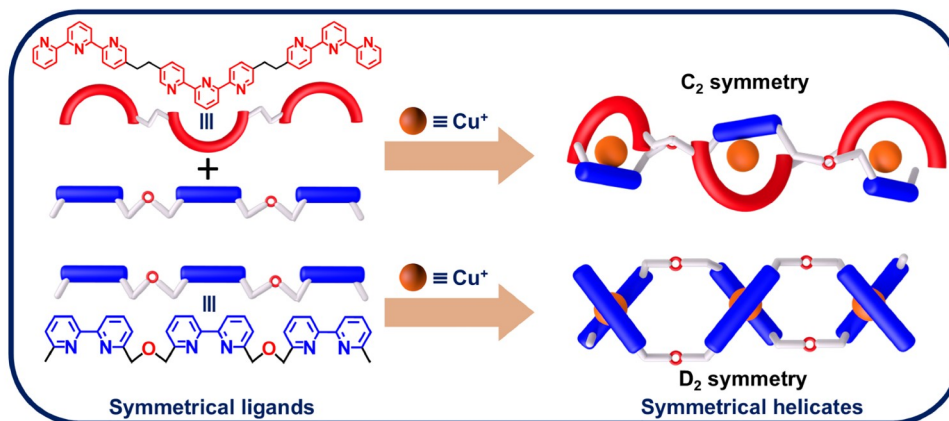
unique double-stranded helicate are required. Moreover, almost all reported conventional double-stranded helical structures constructed by coordination-driven self-assembly have symmetrical axis with C_2 or D_2 symmetry (Scheme 1A) [7,13].

Among the various ligands, 2,2':6',2''-terpyridine (tpy) has attracted much interest in the fields of supramolecular and coordination chemistry, due to its coordination ability with a variety of metal ions and the potential applications of tpy-based complexes [14–24]. The fixed geometry of tpy-M(II)-tpy connectivity has a pseudo-octahedral coordination configuration and ideally has D_{2d} symmetry [14,25], which induced more challenges to assemble into a helicate by tpy-based ligands than ligands based on 2,2'-bipyridine (bpy) or phenanthroline. That is related to the coordination geometry of units $[M(bpy)_2]$ and $[M(tpy)_2]$, where the absolute configuration of the metal ion of $[M(bpy)_2]$ produces right-handed ($\Delta=P$) or left-handed ($\Lambda=M$) helicity of the whole structure. On the contrary, the utilization of $\langle tpy-M-tpy \rangle$ connectivity of multi-terpyridyl ligands provides the desired linear characteristics, which limited the formation of helical metallo-supramolecular structures. Although a number of

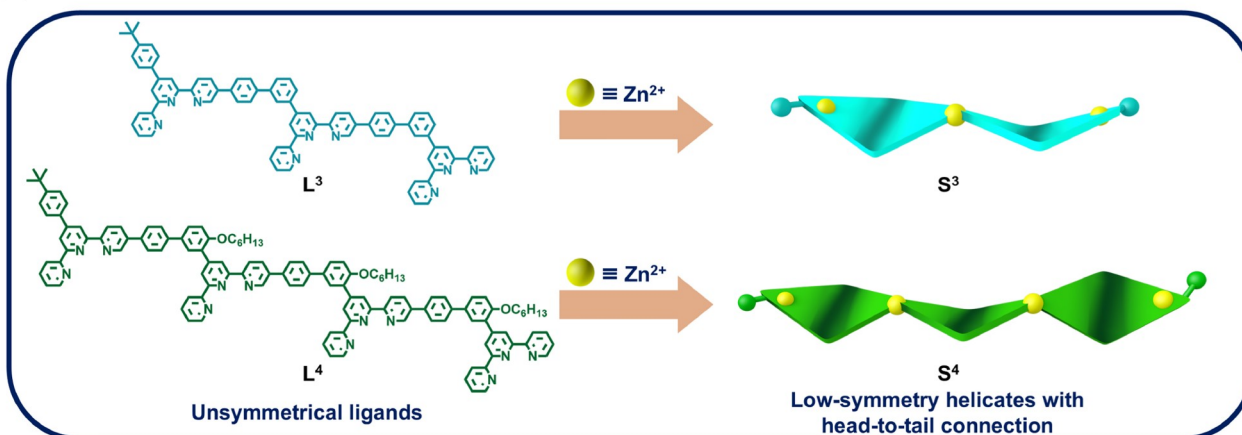
synthetic polymers and oligomers (foldamers) [26,27], which fold into single-handed helical conformations, have been reported to date, only a limited number of structural motifs based on tpy are available for constructing multiple-stranded helical structures [28], in particular, double-stranded helicates [29].

In this study, we constructed a series of extremely low-symmetrical helicates, which are apparently different from the conventional axisymmetric helicates (Scheme 1B). Five ladder-shape ligands (L^2 , L^3 , L^R , L^S , and L^4) with consecutive unsymmetrical modification at 5-position of tpy motifs were designed and synthesized. All ligands can assemble with Zn(II) to form dimeric complexes. The ladder-style tridentate ligand L^3 was first obtained to build a novel helicate S^3 . The single crystal of S^3 revealed its double-stranded helical structure with a particular head-to-tail coordination mode and the racemates with P and M conformations, due to the consecutive unsymmetrical modification of ligand and suitable torsion angle provided by the phenyl group as linkers. To achieve optically pure helicates, the chiral 2,6-bis(oxazolinyl)pyridine (PyBox) group was introduced at the tail of tridentate ligand and achieved

(A) Previous work



(B) This work



Scheme 1 (Color online) (A) Two symmetrical double-stranded helicates in previous work. (B) The extremely low-symmetry double-stranded helicates in this work based on tpy ligands with consecutive unsymmetrical modification (the alkoxy chains of S^4 were removed for simplicity).

chiral resolution of racemic species. Then, a more-complicated tetradentate ligand **L**⁴ and complex **S**⁴ were successfully obtained. Moreover, the self-assembly of mixed ladder-pattern ligands **L**², **L**³, **L**⁴, and traditional tpy **L**¹ with Zn(II) showed excellent narcissistic self-sorting behavior.

2 Results and discussion

The ladder-style tridentate ligand **L**³ (Scheme 1B) was designed to build a novel helicate. In our design, both the modification at 5-position of tpy motifs and the linkers employed phenyl groups played critical roles. The phenyl groups between tpy groups provided sufficiently rigid to both avoid chelation of one ligand around a single metal, and promote mechanical coupling between adjacent stereogenic coordination sphere. Ligand **L**³ was synthesized by Kröhnke and Suzuki coupling reactions (Scheme S1, Supporting Information online). Further, complex **S**³ was obtained by assembling **L**³ with Zn(NO₃)₂·6H₂O in a precise stoichiometric ratio of 2:3 in a mixed solvent of CHCl₃ and MeOH (1:3, v/v) at 50 °C for 8 h, followed by the addition of excess NH₄PF₆ to give a white powder in quantitative yield (97%).

The ¹H nuclear magnetic resonance (NMR) spectra of ligand **L**³ and complex **S**³ are shown in Figure 1A. Given the unsymmetrical nature of **L**³, eight sets of expected pyridines signals attributed to the different tpy moieties were confirmed by ¹H NMR spectra and 2D-correlation spectroscopy (COSY) results (Figures S1 and S5, Supporting Information online). Compared with ligand **L**³, the complex **S**³ exhibited a more complicated signal, due to the enhanced steric hindrance effect caused after coordination, suggesting the formation of complex. After coordination with Zn(II), five significant upfield shifts for protons of 6, 6', a6, a6', and b6 were clearly observed with Δδ≈0.8, 1.2, 0.7, 1.1, and 0.9 ppm, respectively, due to the electron shielding effect [30]. While other protons showed downfield shifts, such as the signal of protons of d, d', e, e', and f, due to the reduced electron density after coordination. In addition, the restrictive structure induced the split of NMR signals. For example, the signal of the tpy-*H*^{a6} split into two sets of peaks, which indicates their rotation restriction caused by the altered chemical environment [31].

Electrospray ionization mass spectrometry (ESI-MS) and traveling-wave ion mobility-mass spectrometry (TWIM-MS) were utilized to further characterize the molecular composition and structural information. In ESI-MS spectrum (Figure 1B), one prominent set of peaks with charge states from 3+ to 5+ was observed, due to the loss of a different number of PF₆⁻. The experimental *m/z* values for each peak were perfectly consistent with the theoretical ones derived from the assembled composition of dimeric **S**³ with a molecular weight of 3,404.5 Da. The TWIM-MS experiments

afforded additional structural evidence, in which a series of consecutive charge signal distributions ranging from 3+ to 5+ derived from complex **S**³ with a narrow drift time were discernible (Figure 1C), indicating the formation of discrete structure without other isomers or structural conformers. These results supported the quantitative formation of a single-component structure **S**³.

Following this result, a tetradentate ligand **L**⁴ with triple consecutive unsymmetrical modification, which has one more meta modification and adopts a similar synthetic route as **L**³ (Scheme S2, Supporting Information online), was synthesized. It is worth mentioning that alkoxy chains were introduced in **L**⁴ to improve the poor solubility (Figure 1), but the alkoxy chains were removed from the cartoon diagram for simplicity (Scheme 1B). Complex **S**⁴ was obtained by the assembly of **L**⁴ and Zn(NO₃)₂·6H₂O in a precise stoichiometric ratio of 2:4, and a similar treatment for **S**³. The ¹H NMR spectra of ligand **L**⁴ and complex **S**⁴ are shown in Figure 1D. Compared with tridentate ligand **L**³, the ¹H NMR spectrum of tetradentate ligand **L**⁴ exhibited a more complicated signal. As many as eleven kinds of expected pyridines signals attributed to four terpyridine moieties were confirmed by ¹H NMR spectra and 2D-COSY results (Figures S12 and S15). For the ¹H NMR spectrum of **S**⁴, seven significant upfield shifts for protons of 6, 6', a6, a6', b6, b6', and c6, due to the electron shielding effect similar to **S**³, indicating the formation of complex **S**⁴.

In ESI-MS spectrum, one prominent set of peaks with charge states from 4+ to 8+ was observed, due to the loss of a different number of PF₆⁻ (Figure 1E). The experimental *m/z* values for each peak were consistent with the theoretical ones derived from the assembled composition of dimeric **S**⁴ with a molecular weight of 5,045.19 Da. The TWIM-MS showed a series of consecutive charge signal distributions ranging from 4+ to 7+ with a narrow drift time (Figure 1F), indicating the formation of a single-component structure **S**⁴ and no other isomers or structural conformers exist.

Although dimeric complexes **S**³ and **S**⁴ were characterized by NMR, ESI-MS and TWIM-MS, detailed structural information about molecular configuration was still urgently needed. The single-crystal characterization has been proven to be the most powerful evidence to confirm the absolute configuration of molecules. The single crystal of **S**³ suitable for X-ray diffraction was successfully obtained by slow vapor diffusion of 1,4-dioxane into an acetonitrile solution of **S**³. As expected, X-ray crystallographic analysis showed that two ligands convolute around two Zn(II) to form a dimer with the head-to-tail coordination mode benefited from the unsymmetrical design of ligand. The helicate **S**³ crystallized into the space group *P*2₁/*c*. Notably, a degree of deformation at the phenyl groups as linker in complex **S**³ provides a torsion angle, possibly to overcome molecular tension of the ligand chain after the coordination, finally resulting in a

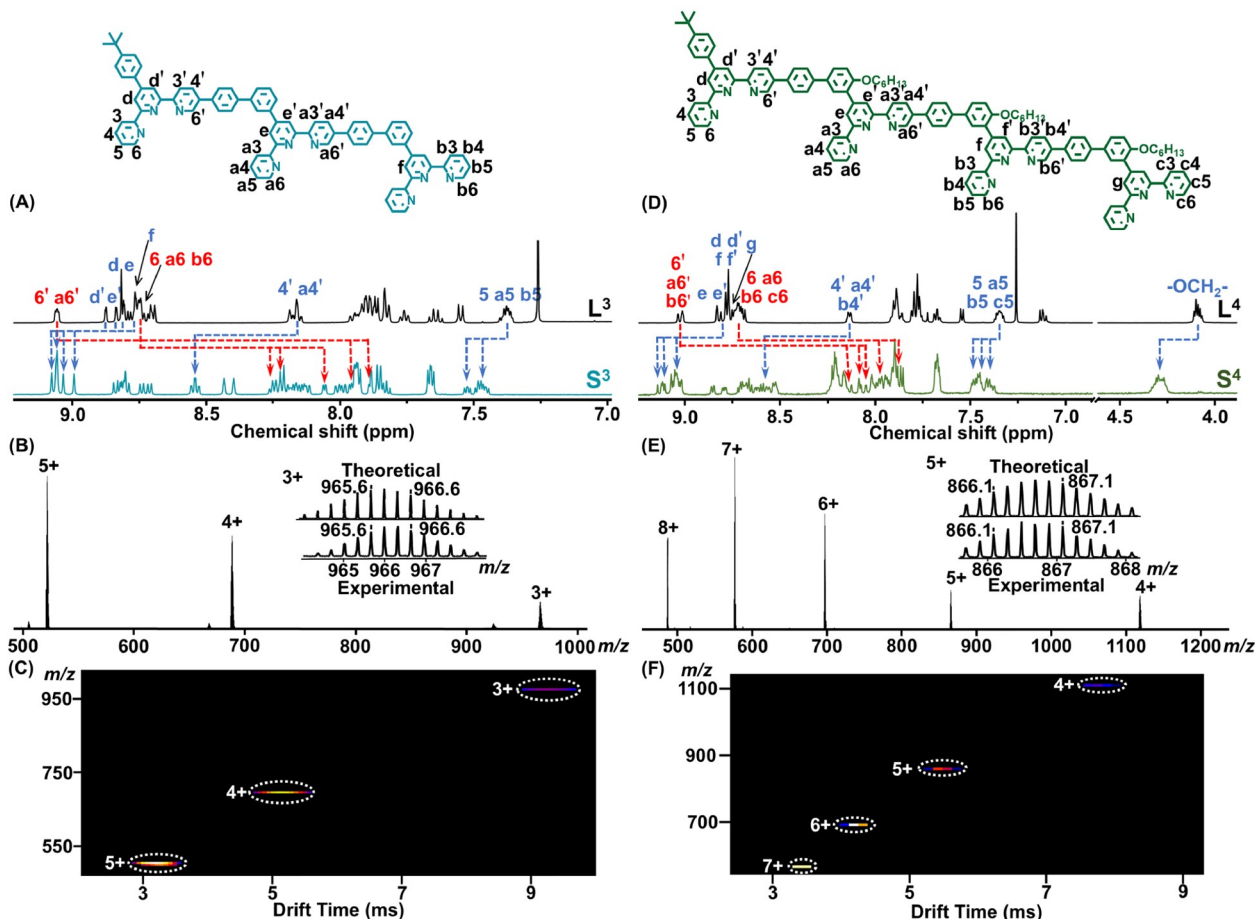


Figure 1 (Color online) (A) 1H NMR spectra (600 MHz, 300 K) of ligand L^3 in $CDCl_3$ and complex S^3 in CD_3CN . (B) ESI-MS spectrum of S^3 (inset: theoretical and experimental isotopic patterns for 3+ moiety). (C) TWIM-MS plot (m/z vs. drift time) of S^3 . (D) 1H NMR spectra (600 MHz, 300 K) of ligand L^4 in $CDCl_3$ and S^4 in CD_3CN . (E) ESI-MS spectrum of S^4 (inset: theoretical and experimental isotopic patterns for 5+ moiety) and (F) TWIM-MS plot (m/z vs. drift time) of S^4 .

double-stranded helical shape for S^3 . More importantly, the single crystal of S^3 exhibited irregularity and unsymmetry which is quite different from traditional helicates [7,13]. The three metal-ion centers in S^3 are not on the straight line and the upper and lower helical ring separated by the intermediate metal are oriented differently, in which the torsion angle between the upper and lower helical ring is 32.7° , making the entire helicate structure irregular and extremely low-symmetry. Moreover, in the single crystal, we find two conformations of the double-stranded helicate S^3 , P - S^3 and M - S^3 , which are perfectly mirror images, making S^3 a racemate (Figure 2A). In the crystal stacking diagram, a layer of P - S^3 and a layer of M - S^3 are stacked on each other, thus leading to the crystals stably stacked. The P - S^3 or M - S^3 of each layer is parallel in the lateral direction (Figure 2B, C). All attempts to grow X-ray-quality single crystals of S^4 have been proven to be unsuccessful, mainly attributed to the following two reasons: one is the increased structural complexity compared with S^3 , and the other is the presence of multiple alkyl chains. Density functional theory (DFT) cal-

culations were used to further understand the architectural features of S^4 (Figure 2D). The configuration of helicate S^4 is even more distorted and the size of helicate S^4 is larger than S^3 due to the additional tpy coordination motif, which increases the spatial constraint. Similar to S^3 , the deformation at the phenyl groups provides a torsion angle, resulting in a double-stranded helical shape for S^4 as well. Four metal centers divide the S^4 into three sailboat bottom-style rings facing different directions (S^4 in Scheme 1B), making the entire structure of helicate S^4 irregular and extremely low-symmetry.

The successful construction of double-stranded helical structures encouraged us to try further chiral resolution. A general strategy to influence the stereochemistry of self-assembled systems is the introduction of additional stereocenters, in the form of either enantiopure building blocks [32–34], chiral ancillary ligands [35,36], chiral guests [37], or counterions [38]. Unfortunately, many attempts to isolate the enantiomers of S^3 have been proven to be unsuccessful to date, such as using chiral column chromatography to sepa-

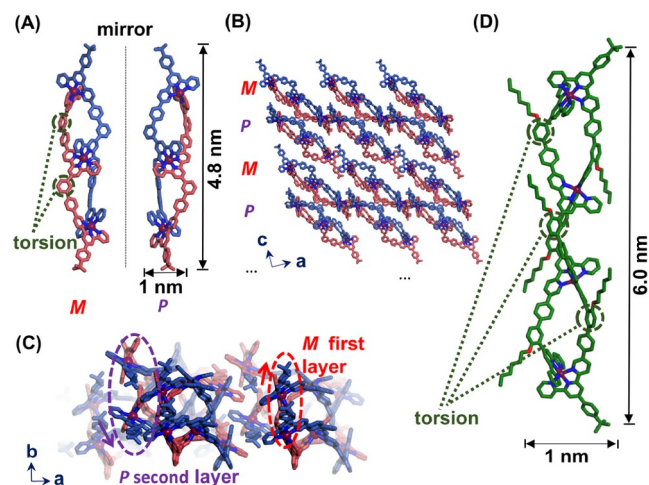


Figure 2 (Color online) (A) The two spatial conformations in the single crystal of S^3 . Crystal packing structure of S^3 in a (B) front view and (C) top view. (D) DFT-optimized structure of S^4 .

rate enantiomers or using chiral auxiliaries to induce chirality. Considering a similar structure and coordination pattern to tpy, we tried to introduce the chiral PyBox group into L^3 to realize chiral resolution (Figure 3A). Detailed synthetic routes of chiral tpy ligands $L^{R/S}$ (Scheme S3) are similar to L^3 . S^S and S^R were obtained by assembling these two optically pure ligands L^S and L^R with $Zn(NO_3)_2 \cdot 6H_2O$ in a precise stoichiometric ratio of 2:3 in a mixed solvent of $CHCl_3$ and MeOH (1:3, v/v) at 50 °C for 8 h, respectively, followed by the addition of excess NH_4PF_6 to give a white powder in quantitative yield (95% for S^S and 93% for S^R). The structural information of S^S and S^R were fully characterized by NMR (Figures S30–S35), ESI-MS and TWIM-MS (Figure S44).

To assess the impact of the incorporated chiral group in these ligands, circular dichroism (CD) spectra of ligands and complexes were further measured. The expected mirror-imaged CD signals were observed for both PyBox group (Figure S47) and $S^{R/S}$ (Figure 3B). The CD spectrum of chiral PyBox group displayed pronounced and well-developed Cotton effects at 220 and 300 nm. The CD spectra of complexes S^R and S^S showed additional Cotton effects between 300 and 450 nm, with highest absorption unsymmetrical factors (g_{abs}) (centered at ~ 385 nm, $|g_{abs}| = 1.09 \times 10^{-3}$) in the whole CD spectra, suggesting the existence of the so-called domino effect [39], which transmitted the chirality of the chiral group to the whole double-stranded helical structure. The above experiment can prove that we have successfully isolated the enantiomers.

To verify whether multiple ligands with different lengths have high self-sorting behavior, we further synthesized the didentate ligand L^2 (Scheme S4 and Figures S36–S39). L^2 , L^3 , L^4 , and common tpy L^1 [40] were mixed together in an equimolar ratio with the corresponding amount of

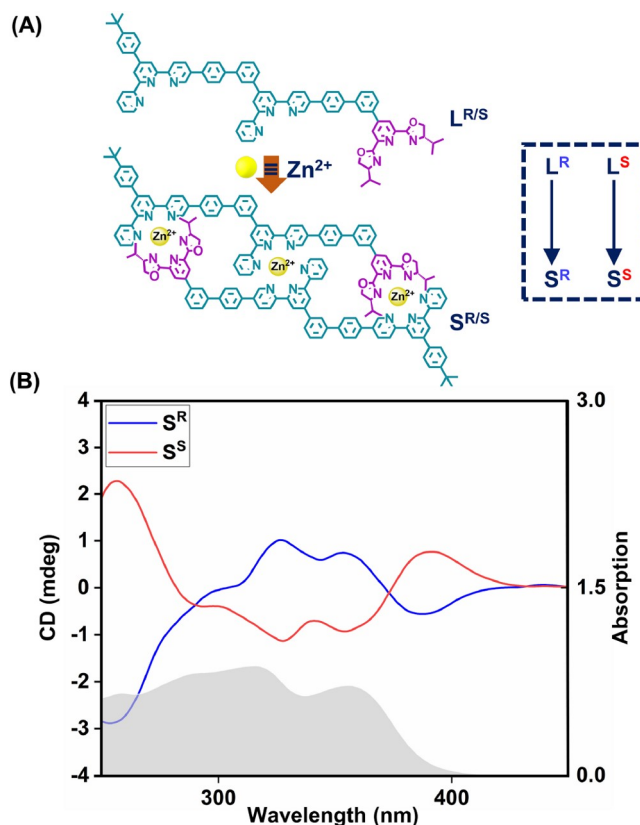


Figure 3 (Color online) (A) The diagram of respective assembling of enantiomers L^R and L^S with $Zn(II)$. (B) CD spectra ($c=10 \mu M$, above) of enantiomers S^R (blue line), S^S (red line) in CH_3CN and absorption spectra ($c=10 \mu M$, bottom, gray shadow).

$Zn(II)$ at 50 °C overnight (Figure 4A). ESI-MS spectrum clearly illustrated four sets of signals for the narcissistic self-sorting complexes S^1 , S^2 , S^3 , and S^4 without statistical mixture (Figure 4B), indicating that these ligands with different consecutive unsymmetrical modification have ideal self-sorting properties in a pluralistic system. As expectation, in accordance with the maximum occupancy of coordination sites proposed by Lehn [41], L^1 – L^4 tend to narcissistically coordinate to form the most energetically favorable structure.

3 Conclusions

In summary, we designed and constructed a series of double-stranded helicates by assembling ladder-style ligands with consecutive unsymmetrical modification. The structural information of ligands and metallo-complexes was confirmed by NMR (1H , ^{13}C , and 2D COSY), ESI-MS, TWIM-MS and X-ray crystallography. The single crystal of S^3 showed that the connection points between different tpy motifs are prone to bending, resulting in a double-stranded helical structure of S^3 . In addition, three metal-ion centers in S^3 are not on the

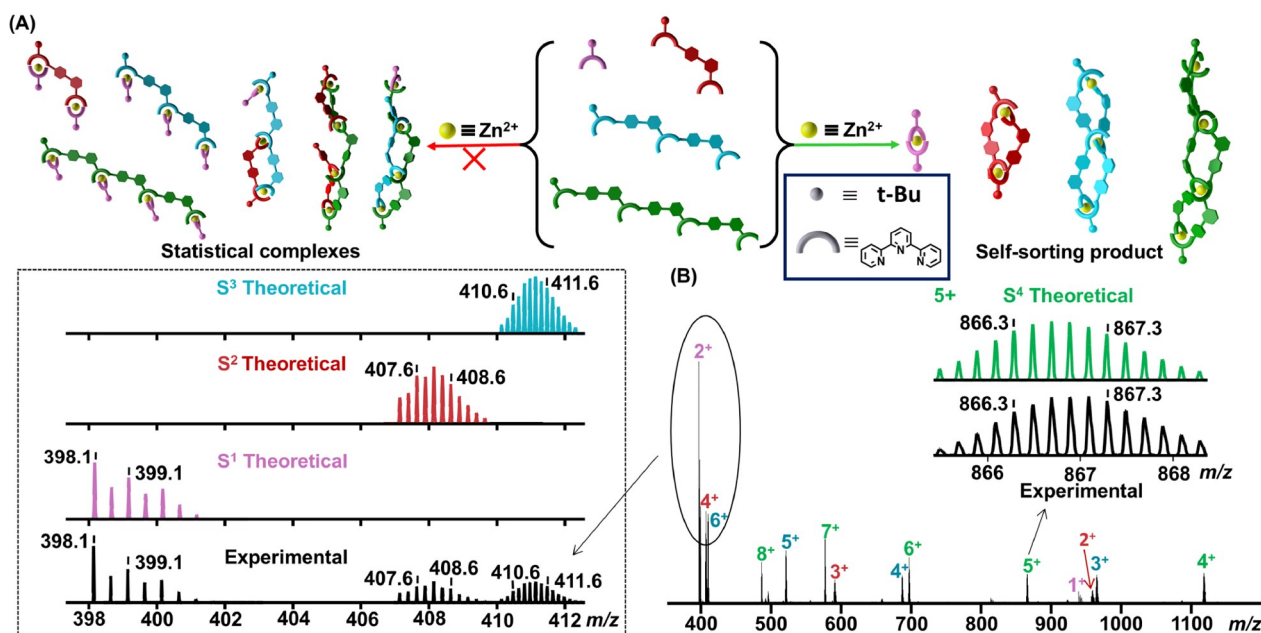


Figure 4 (Color online) The narcissistic self-sorting behavior of L^1 , L^2 , L^3 , and L^4 . (A) Schematic representation for the narcissistic self-sorting process. (B) ESI-MS of the assembly of mixed L^1 , L^2 , L^3 , and L^4 in an equimolar ratio with the corresponding amount of Zn(II).

straight line and the upper and lower helical ring separated by the intermediate metal are oriented differently, in which the torsion angle between the upper and lower helical ring was 32.7° , making the entire structure of helicate irregular and extremely low-symmetry. Further, we introduced the chiral PyBox group in the design of tridentate ligand to obtain two chiral ligands L^R and L^S , and achieve the purpose of chiral resolution of racemic species. Notably, four different complexes S^2 , S^3 , S^4 , and S^1 can be assembled by mixed ladder-pattern ligands L^2 , L^3 , L^4 , and common tpy L^1 with Zn(II) in one pot, indicating the significant narcissistic self-sorting behavior. This work not only provided a new method for preparing helicates, but also developed a new coordination motif for self-recognition, which would provide guidance for the construction of more complex structures in the future.

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Conflict of interest The authors declare no conflict of interest.

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